# Ryan Cory-Wright

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## Academic Appointments

**Imperial College London,** Imperial Business School — *London, UK*
Assistant Professor of Analytics and Operations — *July 2023-present*

**IBM Research** — *Cambridge, MA*
Herman Goldstine Postdoctoral Fellow — *July 2022-June 2023*

## Research Interests

- Optimization: integer, matrix, and tensor; semidefinite and sum-of-squares relaxations; approximation algorithms
- Machine learning and AI: AI for scientific discovery; interpretability; cross-validation; large language models
- Applications: electricity market design and pricing; decarbonization and sustainable operations

## Education

**Massachusetts Institute of Technology,** Operations Research Center — *Cambridge, MA*
Ph.D. in Operations Research — *September 2017-May 2022*
Advisor: Dimitris Bertsimas | Thesis: Integer and Matrix Optimization: A Nonlinear Approach

**University of Auckland,** Faculty of Engineering — *Auckland, New Zealand*
B.E. (1st Class Honors) in Engineering Science — *February 2014-October 2016*
Four-year degree completed in three years via accelerated pathway

## Working Papers

W7. *Stability Regularized Cross-Validation*
**R. Cory-Wright** and A. Gómez, working paper.

W6. *Pricing Discrete and Nonlinear Markets: A Semidefinite Sensitivity Approach*
C. Guo, L. Henderson, **R. Cory-Wright**, and B. Yang, working paper.

W5. *msPCA: An R Package for Sparse PCA With Multiple Components*
**R. Cory-Wright** and J. Pauphilet, working paper.
- R package **msPCA** available on CRAN **[link]**

W4. *A Shrinkage Path Heuristic for Wasserstein Distributionally Robust Optimization*
L. Meng, **R. Cory-Wright**, and W. Wiesemann, submitted.

W3. *Compact Lifted Relaxations for Low-Rank Optimization*
**R. Cory-Wright** and J. Pauphilet, submitted.

W2. *Improved Approximation Algorithms for Orthogonally Constrained Problems Using Semidefinite Optimization*
**R. Cory-Wright** and J. Pauphilet, under review at special issue of **Mathematical Programming B**.
- **Preliminary version accepted at IPCO 2026 [link]**

W1. *Bridging the Gap Between Scientific Laws Derived by AI Systems and Canonical Knowledge via Abductive Inference with AI-Noether*
K. Srivastava, S. Dash, **R. Cory-Wright**, B. Trager, C. Cornelio, and L. Horesh, major revision at **Nature Communications**.

## Journal Papers

J17. *Disjunctive Branch-And-Bound for Certifiably Optimal Low-Rank Matrix Completion*
D. Bertsimas, **R. Cory-Wright**, S. Lo, and J. Pauphilet, minor revision at **INFORMS Journal on Computing**.

J16. *Efficient Cross-Validation for Sparse Linear Regression*
**R. Cory-Wright** and A. Gómez, minor revision at **INFORMS Journal on Computing.**

J15. *Sparse PCA With Multiple Components*
**R. Cory-Wright** and J. Pauphilet, **Operations Research**, Articles in Advance, 2026.
- **First place, INFORMS DMDA Workshop Paper Award (Theoretical Track, 2024)**

J14. *The Need for Verification in Artificial Intelligence-Driven Scientific Discovery*
C. Cornelio, T. Ito, **R. Cory-Wright**, S. Dash, and L. Horesh, **Philosophical Transactions of the Royal Society A**, 384(2317), 2026.

J13. *Decarbonizing OCP*
D. Bertsimas, **R. Cory-Wright**, and V. Digalakis Jr., **Manufacturing & Service Operations Management**, 27(6): 1760-1778, 2025.
- **Finalist, M&SOM practice-based research competition (2023)**
- **Honorable mention, MIT Operations Research Center Student Paper Award (Digalakis, 2023)**
- Featured in Imperial Business news article "Optimising renewables: a model for profitable decarbonisation" [link]

J12. *A Stochastic Benders Decomposition Scheme for Large-Scale Stochastic Network Design*
D. Bertsimas, **R. Cory-Wright**, J. Pauphilet, and P. Petridis, **INFORMS Journal on Computing**, 37(5):1163–1181, 2025.

J11. *Evolving Scientific Discovery by Unifying Data and Background Knowledge with AI Hilbert*
**R. Cory-Wright**, C. Cornelio, S. Dash, B. El Khadir, and L. Horesh, **Nature Communications** 15:5922, 2024.
- **IBM Outstanding Technical Achievement Award (2024)**
- Featured in IBM Research blog "Meet AI Hilbert, a new algorithm for transforming scientific discovery" [link]

J10. *Sparse Plus Low Rank Matrix Decomposition: A Discrete Optimization Approach*
D. Bertsimas, **R. Cory-Wright**, and N. A. G. Johnson, **Journal of Machine Learning Research**, 24(267):1–51, 2023.
- **First place, INFORMS Data Mining Society Student Paper Award (2021)**
- **A. E. Grant Poster Award for Best Algorithm, CAARMS (Johnson, 2022)**

J9. *A New Perspective on Low-Rank Optimization*
D. Bertsimas, **R. Cory-Wright**, and J. Pauphilet, **Mathematical Programming A**, 202(1-2):47–92, 2023.

J8. *Mixed-Projection Conic Optimization: A New Paradigm for Modeling Rank Constraints*
D. Bertsimas, **R. Cory-Wright**, and J. Pauphilet, **Operations Research**, 70(6):3321–3344, 2022.
- **First place, INFORMS George Nicholson Student Paper Competition (2020)**

J7. *A Scalable Algorithm for Sparse Portfolio Selection*
D. Bertsimas and **R. Cory-Wright**, **INFORMS Journal on Computing**, 34(3):1489-1511, 2022.

J6. *Solving Large-Scale Sparse PCA to Certifiable (Near) Optimality*
D. Bertsimas, **R. Cory-Wright**, and J. Pauphilet, **Journal of Machine Learning Research**, 23(13):1-35, 2022.

J5. *A Unified Approach to Mixed-Integer Optimization Problems With Logical Constraints*
D. Bertsimas, **R. Cory-Wright**, and J. Pauphilet, **SIAM Journal on Optimization**, 31(3):2340-2367, 2021.
- **First place, INFORMS Computing Society Student Paper Award (2019)**

J4. *From Predictions to Prescriptions: A Data-Driven Response to COVID-19*
D. Bertsimas, L. Bouissoux, **R. Cory-Wright** et al., **Health Care Management Science**, 24:253-272, 2021.
- **First place, INFORMS Health Applications Society Pierskalla Best Paper Award (2020)**

J3. *On Stochastic Auctions in Risk-Averse Electricity Markets With Uncertain Supply*
**R. Cory-Wright** and G. Zakeri, **Operations Research Letters**, 48(3):376-384, 2020.

J2. *On Polyhedral and Second-Order Cone Decompositions of Semidefinite Optimization Problems*
D. Bertsimas and **R. Cory-Wright**, **Operations Research Letters**, 48(1):78-85, 2020.

J1. *Payment Mechanisms for Electricity Markets With Uncertain Supply*
**R. Cory-Wright**, A. Philpott, and G. Zakeri, **Operations Research Letters**, 46(1):116-121, 2018.
- **First place, Operations Research Society of New Zealand Student Paper Award (2016)**

## Articles in Preparation

P2. *Designing Optimal Matrix Multiplication Algorithms via Low-Rank Tensor Decompositions*
**R. Cory-Wright**, J. Pauphilet, and Y. Wu, in progress.

P1. *Making Optimization Interpretable With Optimal Decision Trees*
D. Keehan, **R. Cory-Wright**, and A. Jacquillat, in progress.

## Books in Preparation

B1. *Integer and Matrix Optimization: A Nonlinear Approach*
D. Bertsimas, **R. Cory-Wright**, and J. Pauphilet, in preparation.

## Selected Awards

Note: * denotes an award won by a collaborator for coauthored work

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| 2024 | First place, **INFORMS DMDA Workshop Paper Award**, Theoretical Track |
| 2023 | Honorable mention, **Student Paper Award**, MIT ORC (Digalakis Jr.*) |
| 2023 | Finalist, **Practice-Based Research Competition**, M&SOM Society |
| 2022 | **A. E. Grant Poster Award for Best Algorithm**, CAARMS (Johnson*) |
| 2022 | **IBM Herman Goldstine Fellowship**, IBM Department of Mathematical Sciences |
| 2021 | First place, **Student Paper Award**, INFORMS Data Mining Society |
| 2020 | First place, **George Nicholson Student Paper Competition**, INFORMS |
| 2020 | First place, **Pierskalla Best Paper Award**, INFORMS Health Applications Society |
| 2019 | First place, **Student Paper Award**, INFORMS Computing Society |
| 2017 | **Senior Scholar Award** (top of cohort), University of Auckland |
| 2016 | First place, **Student Paper Award**, Operations Research Society of New Zealand |

## Research Funding

Under review:

1. UKRI EPSRC New Investigator Award, *Matrix Multiplication Algorithms*, £626,418 (full economic cost), Principal Investigator, submitted November 2025

## Teaching

### Imperial

Fundamentals of Python (MSc AI Applications and Innovation) — *Imperial-X*
Course Creator and Instructor — *Fall 2025, 2026*

Introduction to Machine Learning in Python (MSc AI Applications and Innovation) — *Imperial-X*
Course Creator and Instructor — *Fall 2024*

Decision Making Under Uncertainty (PhD) — *Imperial Business School*
Course Creator and Instructor — *Spring 2024, 2025, 2026, 2027*

Data Structures and Algorithms (undergraduate) — *Imperial Business School*
Course Creator and Instructor — *Spring 2024, 2025, 2026, 2027*

Optimization and Decision Models (Online MSc Business Analytics) — *Imperial Business School*
Instructor — *Spring 2024*

### MIT

15.095 Machine Learning Under a Modern Optimization Lens (MBAn/PhD) — *MIT*
Head Teaching Assistant — *Fall 2019, 2021*

15.071 The Analytics Edge (MBA) — *MIT*
Head Teaching Assistant — *Fall 2020*

15.093 Optimization Methods (MSc/PhD) — *MIT*
Teaching Assistant — *Fall 2018*

Kaufman Teaching Certificate Program — *MIT Teaching and Learning Lab*
Participant, eight practice-based workshops on teaching effectiveness — *Fall 2021*

## Student Advising

### Postdoctoral Fellows

1. Dominic Keehan, *ICRF Postdoctoral Fellow at Imperial Business School* (co-mentored with Wolfram Wiesemann, research at the intersection of optimization and AI).

### Doctoral Students

1. Lingjun Meng, *PhD student at Imperial Business School* (co-advised with Wolfram Wiesemann, research on optimization under uncertainty with applications to sustainability).

## Oral Presentations

### Invited Presentations at Academic Institutions and Single-Track Workshops

*Improved Approximation Algorithms for Orthogonally Constrained Problems Using Semidefinite Optimization*

- Princeton ORFE — *2026*
- Imperial Analytics and Operations (internal seminar) — *2026*
- CMU Tepper OR — *2025*
- Michigan IOE — *2025*
- IBM Yorktown Heights — *2025*
- Workshop on Information Learning — *2025*

*Sparse PCA With Multiple Components*

- Cornell ORIE — *2025*
- Northwestern IEMS — *2025*
- Imperial-X — *2024*

*Evolving Scientific Discovery by Unifying Data and Background Knowledge with AI Hilbert*

- Cornell Tech — *2025*
- The Alan Turing Institute — *2024*
- Summer Workshop on Innovations in Management Science — *2024*

*Optimal Low-Rank Matrix Completion: Semidefinite Relaxations and Eigenvector Disjunctions*

- Toronto Rotman Young Scholar Seminar Series — *2023*
- Imperial College London Control and Optimization — *2023*
- Mixed Integer Programming Workshop — *2023*

*A New Perspective on Low-Rank Optimization*

- IBM Yorktown Heights — *2024*
- Lehigh ISE — *2022*

*Mixed-Projection Conic Optimization: A New Paradigm for Modeling Rank Constraints*

- IBM Yorktown Heights — *2022*
- Rice CMOR — *2022*
- CMU Tepper OR — *2022*
- USC Viterbi ISE — *2022*
- Georgia Tech ISyE — *2022*
- Johns Hopkins Carey OM — *2022*
- Princeton ORFE — *2022*
- Imperial Analytics and Operations — *2021*
- University of Auckland Engineering Science — *2020*

### Invited Presentations at Companies

*The Future of Artificial Intelligence*

- South Port New Zealand Board of Directors Meeting *2024*

## Other Academic and Industry Experience

Collaborations with organizations: Analytics for a Better World (2025-present), OCP (2021-22), CIBC (2017-20).

University of Auckland, Department of Engineering Science *Auckland, New Zealand*
Research Assistant *December 2016-July 2017*

SUEZ Smart Solutions *Auckland, New Zealand*
Assistant Optimization Engineer *December 2014-February 2016*

## Activities and Service

### Organizing seminars and workshops

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| 2027 | Organizing committee member, Mixed Integer Programming Workshop |
| 2026 | Cluster chair, INFORMS Optimization Society Conference |
| 2024-27 | Co-initiator and co-organizer, London Operations Research Day (LORD) [web link] 2024 LORD at LBS, 95 registered participants; 2025 LORD at Oxford, 120 registered participants; 2026 LORD at Imperial, 100 registered participants; 2027 LORD at UCL |
| 2019- | Session chair, INFORMS Annual Meeting, ICCOPT, IOS, ISMP, and SIOPT |
| 2019 | Co-organizer, MIT ORC student seminar series |

### External

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| 2025 | Committee member, INFORMS Computing Society Student Paper Award |
| 2024-26 | Judge, M&SOM Student Paper Competition |
| 2017- | Member, INFORMS (Main, Computing Society, Optimization Society) Member, Mathematical Optimization Society |

### Imperial

Note: IB denotes service for Imperial Business School, ICL denotes service for Imperial College London

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| 2026- | Seminar co-organizer, Analytics and Operations group | *IB* |
| 2025-26 | Coordinator, Analytics and Operations faculty hiring | *IB* |
| 2025- | Member, AI and Education Committee | *IB* |
| 2025- | Member, Economics of AI initiative | *ICL* |
| 2024-25 | PhD early/late-stage assessments, Yanwei Sun, Zhongze Cai, Fupeng Sun | *IB* |

### Peer Review

Reviewer for academic journals: *Operations Research (OR), Management Science (MS), Manufacturing & Service Operations Management (M&SOM), Mathematical Programming (MAPR), Journal of Machine Learning Research (JMLR), Mathematics of Operations Research (MOOR), Foundations of Computational Mathematics (FOCM), INFORMS Journal on Computing (IJOC), INFORMS Journal on Optimization (IJOO), SIAM Journal on Optimization (SIOPT), Transportation Science (TS), SIAM Journal on Matrix Analysis and Applications (SIMAX), SIAM Journal on Mathematics of Data Science (SIMODS), Operations Research Letters (ORL), European Journal of Operational Research (EJOR).*

- 2024 Meritorious Reviewer Award, INFORMS Journal on Computing.
- Reviewing record: ORCID [link] | Web of Science [link]

Reviewer for extended abstracts/papers at conferences: EC 2026, ICML 2026, IPCO 2026, M&SOM 2025, IPCO 2024.

- 2026 gold reviewer (top 25%), ICML.